

FIELD THEORY PART-II

Phase 2: Curl Mastery

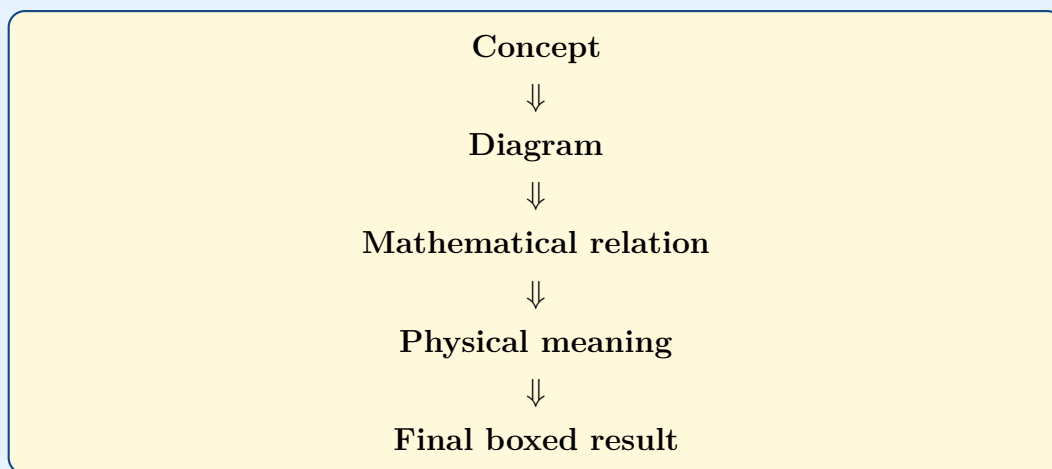
Definition, derivation, diagram and physical significance

Academic Article Style Notes for Semester Preparation

How to Use This Phase

This phase is made for semester preparation from a very basic level. Read the concept first, then understand the diagram, then learn the derivation pattern, and finally practise the exam-style writing block.

The recommended writing order is:



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1 Aim of Phase 2

Main Goal

Curl is one of the most repeated topics in Field Theory Part-II. This phase prepares a complete semester-style answer on curl from definition to physical significance.

2 Definition of Curl

Fundamental Definition

For a vector field $\mathbf{A}$, the component of curl along unit normal $\hat{\mathbf{n}}$ is defined as:

$$(\nabla \times \mathbf{A}) \cdot \hat{\mathbf{n}} = \lim_{\Delta S \rightarrow 0} \frac{1}{\Delta S} \oint_C \mathbf{A} \cdot d\mathbf{l}$$

Here C is the boundary of small surface ΔS .

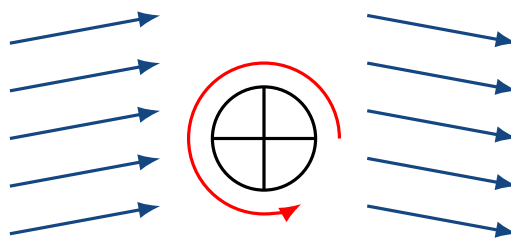
3 Physical Meaning Before Mathematics

Paddle-Wheel Idea

Curl measures local rotation. If a small paddle wheel is placed in a field and it rotates, then curl exists at that point.

If curl is zero, the field is irrotational.

Curl means local rotation or circulation



Paddle wheel picture

Figure 1: Physical significance of curl.

4 Derivation of Curl in Cartesian Coordinates

Starting Vector

Let

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}.$$

To derive the z -component of curl, take a small rectangular loop in the xy -plane.

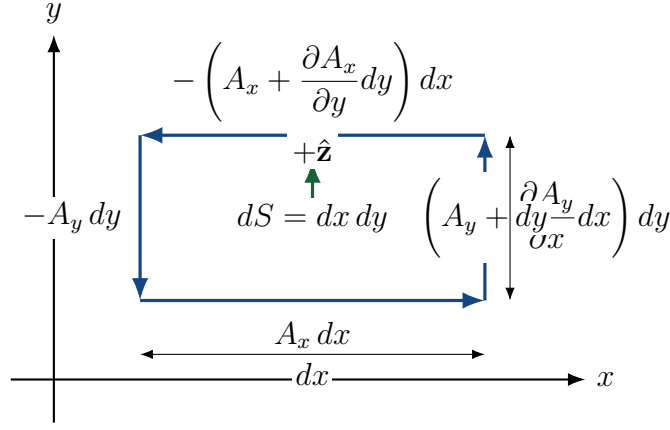


Figure 2: Small rectangular path for deriving the z -component of curl.

Line Integral Around the Four Sides

The circulation is:

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = A_x dx + \left(A_y + \frac{\partial A_y}{\partial x} dx \right) dy - \left(A_x + \frac{\partial A_x}{\partial y} dy \right) dx - A_y dy.$$

After cancellation:

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) dx dy.$$

Since $\Delta S = dx dy$:

$$(\nabla \times \mathbf{A})_z = \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}$$

Other Components

Similarly:

$$(\nabla \times \mathbf{A})_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}$$

and

$$(\nabla \times \mathbf{A})_y = \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}$$

Final Cartesian Expression

$$\nabla \times \mathbf{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{\mathbf{z}}$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

5 Exam Writing Template

How to Write in Semester

Write in this order:

1. Definition of curl.
2. Draw small rectangular loop in xy -plane.
3. Write line integral contributions of all four sides.
4. Obtain z -component.
5. Write other components by cyclic symmetry.
6. Write determinant form.
7. Explain physical meaning.

6 Final Self-Test for This Phase

Instructions

Answer these without looking at the notes. If you cannot answer a point, revise that section again before moving to the next phase.

1. Can I state the main definition of this phase?
2. Can I draw the main diagram neatly without overlap?
3. Can I explain the physical meaning in two or three lines?
4. Can I write the final formula with correct symbols?
5. Can I connect this topic with at least one past-year question pattern?
6. Can I write a semester-style answer with definition, diagram, derivation and final result?

Completion Rule

This phase is complete only when the concept, diagram, formula and answer-writing pattern are all ready.